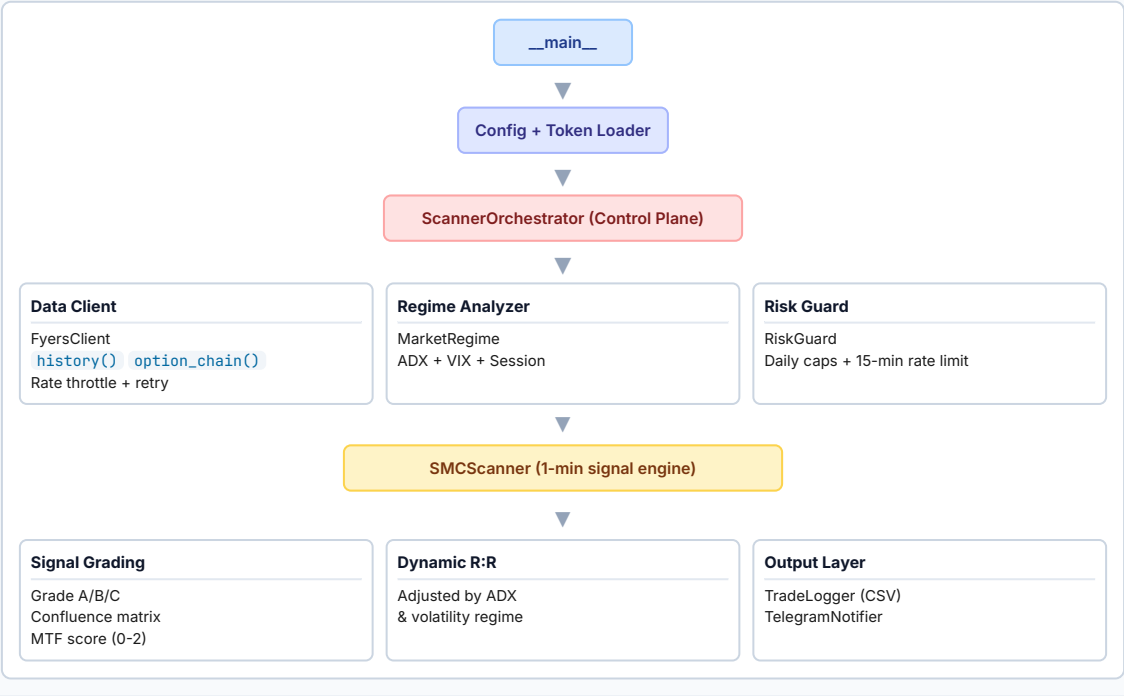


SMC Breakout Scanner v3.0

Architectural Blueprint & Data Flow Diagram

SYSTEM ARCHITECTURE



SCAN CYCLE EXECUTION FLOW

Pre-Flight Checks

- Market open? (09:15 - 15:40 IST)
- NSE holiday check (live API + cache)
- Regime analysis (ADX, VIX, session)
- Risk Guard gate (daily cap, 15-min window)

Data Acquisition

- Fetch NIFTY spot (1-min, 3 days)
- Fetch active futures (cont_flag=1)
- Compute VWAP + EMA20 bias
- Resolve option chain (live or manual)

Signal Generation

- Drop forming candle
- ATR, pivots, OB, FVG detection
- Liquidity sweep + inducement walk
- Displacement filter (body > ATR x mult)
- Trend filter + volume filter

Post-Processing

- HTF confirmation (5-min displacement)
- MTF alignment (3m/5m EMA score)
- Grade assignment (A/B/C)
- Dynamic R:R + suggested qty
- Expiry-day OTM safety filter
- Log + Telegram dispatch

CORE CLASSES & RESPONSIBILITIES

Class	Role
Config	Central parameter store (SMC, risk, infra)
FyersClient	API abstraction, throttling, retry logic
Indicators	ATR, ADX, EMA, VWAP, pivots, OB, FVG
SMCScanner	1-min signal detection engine
MarketRegime	Trend/volatility/session context
RiskGuard	Position sizing, rate limits, R:R
TradeLogger	Thread-safe CSV audit trail
TelegramNotifier	HTML alert dispatch with chunking
SymbolGenerator	Expiry + strike resolution
FuturesHelper	Monthly future symbol + rollover

SIGNAL GRADING MATRIX

Grade	Criteria	Qty
A	Strong close + OB/FVG + volume spike + HTF	3
B	2 of 3 confluences (OB, FVG, volume)	2
C	Basic sweep + displacement only	1

Quantity scales down in extreme VIX or low ADX regimes.

KEY DATA FIELDS (SIGNALRESULT)

Field	Description
symbol	Option symbol (e.g., NSE:NIFTY26O0319800CE)
signal	LONG / SHORT / NO_SIGNAL / NO_DATA
grade	A / B / C confluence grade
htf_confirmed	5-min displacement alignment
mtf_score	0-2 (3m/5m EMA alignment)
regime	strong_trend / trending / ranging
expected_rr	Dynamically adjusted risk:reward
suggested_qty	Lot multiplier based on grade + regime
confluences	Pipe-delimited list of triggers

V3.0 NEW FEATURES

- **NEW** Order Block detection
- **NEW** Fair Value Gap detection
- **NEW** Multi-timeframe alignment (3m/5m)

- **NEW** ADX + India VIX regime engine
- **NEW** Dynamic risk:reward adjustment
- **NEW** Grade A/B/C signal classification
- **NEW** Daily & 15-min rate limits
- **NEW** Expiry-day OTM safety filter
- **NEW** Live NSE holiday API + cache

Generated blueprint for SMC Breakout Scanner v3.0.0 | Landscape A4 | One-page reference